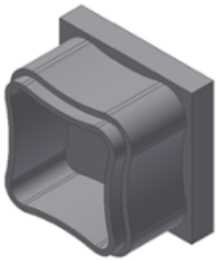
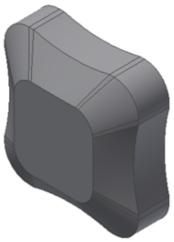
Jeide, Matthew 4/17/26 Period 2

W18: CIM 2.3.7 [PROJECT] Container Design CNC Milling



INTRODUCTION

In this project, you will apply your knowledge and skills about 3D modeling and **G&M codes**. You will generate a CAD model, develop a CAM strategy, verify the strategy, and then possibly machine the part using a CNC Mill. You will complete multi-tool operations using the CNC mill.



EQUIPMENT

CNC Motion Software

The following information should be included on this page and inserted in your E-Portfolio.

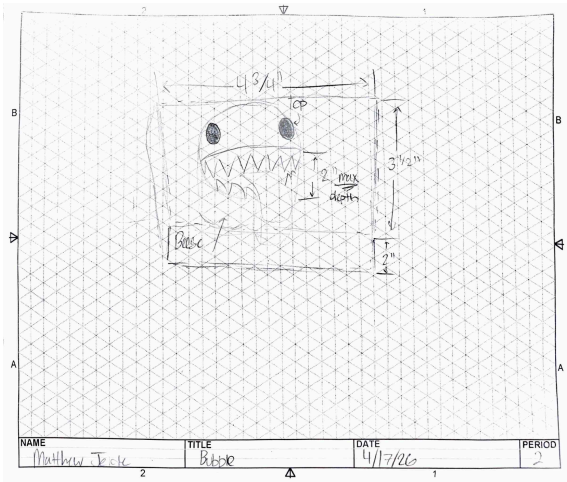
Complete the container design constraints below for Design Strategy 1 or Design Strategy 2 using the information provided by your teacher. Note that Design Strategy 2 produces the base and lid out of the same piece of stock and needs to be separated after milling.

Brainstorm

Sketching the ideas

Brainstorm an idea for a container design using Isometric Paper. In the sketch, show dimensions appropriate to the opening and closing of the container without the lid falling off.

https://drive.google.com/file/d/1A3O86jsbyshdBWhZOEzrQOMx_efiPDSp/view?usp=sharing



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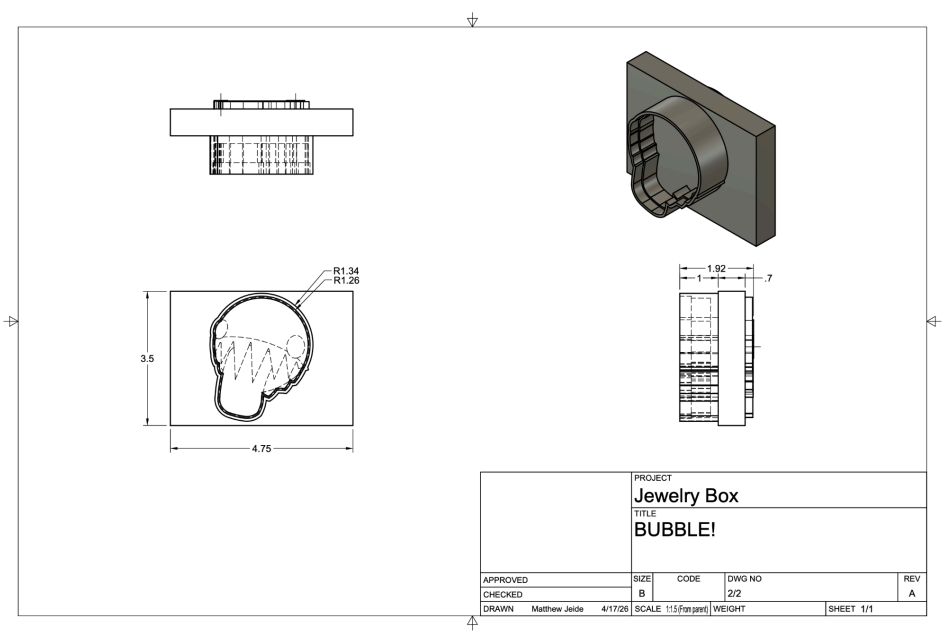
Fusion 360 Design

1. Design and create a unique container base and lid using the constraints and process that follows.
    a. Constraints
        i. CAD model must match the exterior dimensions provided.
        ii. The CAD model must have the clearance fit provided.
        iii. The part machining should take less than 10 minutes.
        iv. The milling operation must require at least one tool change.
    b. Process
        i. Sketch preliminary designs in your PLTW Engineering Notebook.
        ii. Create 3D CAD models of the part base, lid, and assembly.
        iii. Create dimensioned drawings.
        iv. Create a CAM model of the part.
        v. Simulate the machining operation in CNCMotion.
        vi. Produce the physical parts using a mill.

Important Size Constraints:

The Container Design must be created on one single block of size 4 6/8 x 3 ½ x 2 cubic inches.

BOX: Multiview of the Base of the Container fully dimensioned.



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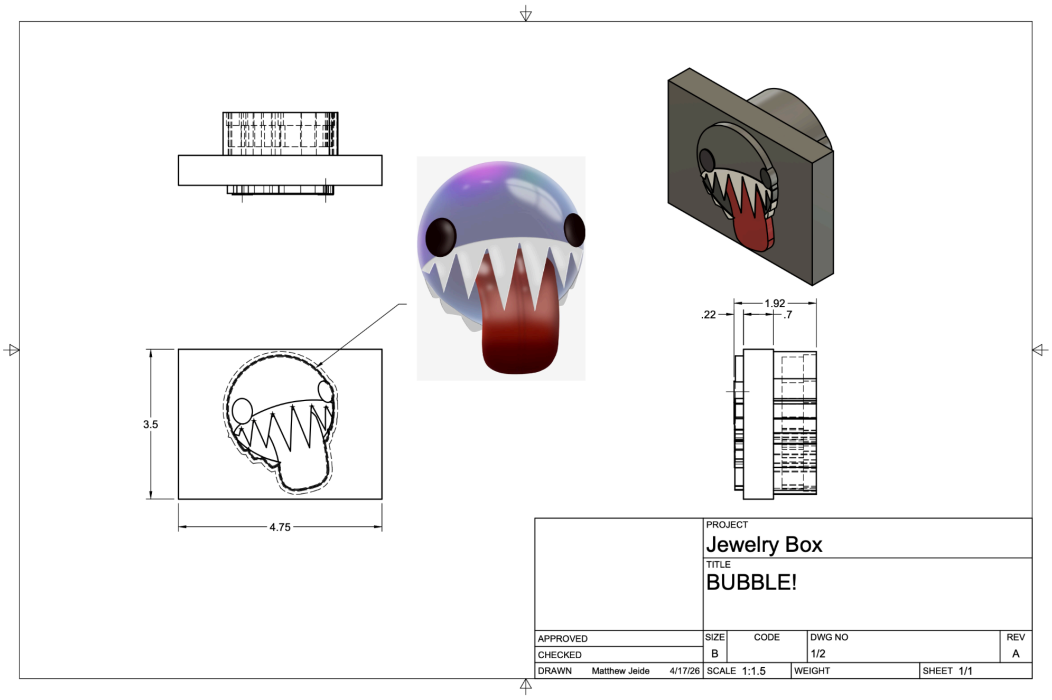
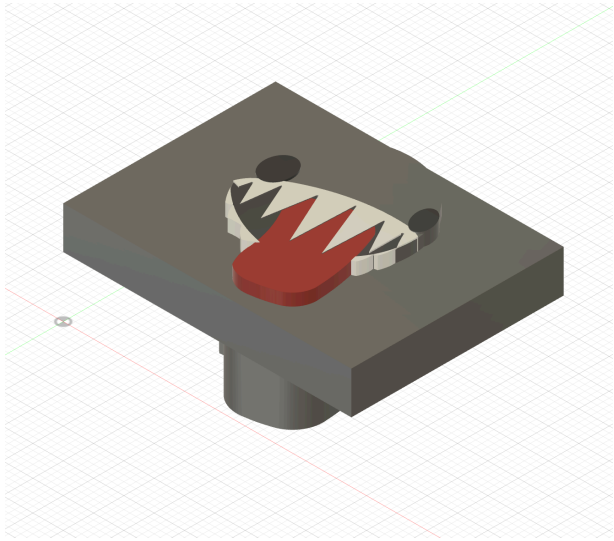
LID: Multiview of the LID fully dimensioned.



Container Design created for Milling on Fusion 360.
~~ASSEMBLY: The top and the bottom of the container fully assembled.~~



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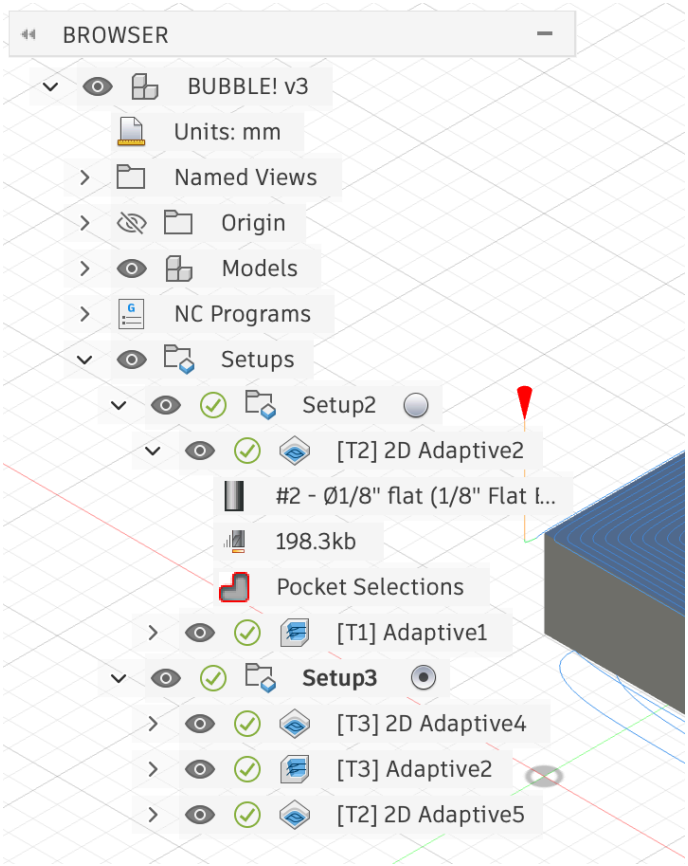
2. **Create the tool paths on Fusion 360 Manufacturing mode.**

NC Code for the Milling Operation on Fusion 360.
https://m-jeide.github.io/eng-portfolio/resources/CIM/Project/CIM%202.3.7/Bubble%20G-Code.pdf#zoom=page-width

3. **Simulate the machining operation in CNCMotion.Verify the CNC Motion Tool 1 has a ⅛ diameter and a Height/Offset of ¼ . The Material Type should be Carbide.**

Insert an image of your Tool Setup here.



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CNC Motion Simulation

Submit a final report containing the following items:

a. Report that uses the CIM Project Report Template.
b. Screen capture of container base and lid models using the following software:
   i. CAD model displayed as an isometric view.
   ii. CAM model showing **toolpaths**.
   iii. CNCMotion simulation with the cycle time displayed.
c. Dimensioned drawings of container base and lid.
d. Completed physical container base and lid.

4. **Insert a video of the milling operation using CNC Motion.**

Video
https://www.youtube.com/watch?v=eb8u8hF1B2Q

Voting on the Best Design

Insert a screenshot of your design using an STL File on IdeaMaker to find out how long it would take to 3D print container design.

Student Design
https://docs.google.com/presentation/d/1ronKDi1Ea_fmyVsuc1ZsZmdpsJnwt3fLwP5IkAaIW3U/edit?usp=sharing **Note for reader: this was a prior years', at the time of the creation of this document, we have yet to be given a current one.**

Portfolio

5. **Insert your portfolio link.**

Portfolio link
https://m-jeide.github.io/eng-portfolio/CIM/CIM%202.3.7

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Conclusions

Use complete sentences to answer the questions below. It is recommended to blue so that I can easily see what you typed.

1. What milling limitations impacted your design?

The flute length of the given tools were at times too short to mill the design without causing errors, more than that using precise tools would lead to longer manufacture times, making the overall design more inefficient to manufacture.

2. What manufacturing processes other than milling could be used to fabricate your container?

Fused Deposition Modeling, or 3D printing.

3. What advice would you like have had prior to starting this project?

To have accounted for the reversal of my design.